AI-Driven Signature Forgery Detection: A Novel Deep Learning-Based Approach with CALSigNet

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***Abstract*—The increased usage of forged signatures is posing a major threat to the legitimacy of legal and financial documents, which requires effective automated detection systems. This paper introduces CALSigNet, a deep learning model using CNN, Attention, and LSTM to improve signature forgery detection accuracy. The model employs CNN for spatial feature extraction, Attention to analyse important signature areas, and LSTM to identify sequential patterns, overcoming the limitations of conventional approaches that tend to neglect both spatial and temporal dynamics. With a large dataset of signature images, the proposed model was tested through intensive training, 5-fold cross-validation, and comparison with baseline CNN models. CALSigNet attained a high-test accuracy of 99.60%, which surpassed that of a Basic CNN (97.60%) and a ResNet-like CNN (98.40%), with a cross-validation mean accuracy of 95.44% (±0.0086). The results highlighted CALSigNet's potential as a scalable, accurate tool for signature verification in forensic science, improving the reliability of legal and financial document authentication.**

Keywords—***AI, Deep Learning, Signature Detection, Image Classification, CALSigNet, CNN, Attention, LSTM, Forged Signature, Prediction.***

I. INTRODUCTION

Today, in the technological domain, Artificial Intelligence (AI) has evolved as a milestone. It has profoundly shaped the future across different domains to unprecedented levels of automation, accuracy, and efficiency. AI has revolutionized industries by processing Big Data, making smart decisions, and detecting patterns. It can also be an innovation tool in forensic science [1], [2]. AI has found its way in the domain of forensic science to provide new solutions for crime scene analysis, document verification, and facial recognition [3]. Investigators now have the ability of rapid analysis, with accuracy, so that the evidence proves better in court sessions. AI has brought an advancement to forensic science, and it has lifted human capability to new heights. It is countering the rising difficulty of modern-day crimes, which can no longer be overcome by traditional methods [4], [5] .

The detection of forged signatures presents one such core challenge for forensic science since it is such a common challenge with profound impacts on legal and financial frameworks. Forgery of signatures threatens the credibility of contracts, wills, and financial transactions, resulting in frauds, disputes, and large economic loss [6]. According to the Association of Certified Fraud Examiners (ACFE), in 2022 [7], billions of dollars of loss is caused by document fraud, such as forged signatures, globally every year. It is important to detect forged signatures because methods used for detection manually by experts can be time-consuming, subjective, and prone to human error, especially in detecting high-quality forgeries [8]. Therefore, it is necessary to develop automated detection systems capable of providing reliability, scalability and objectivity especially in legal and financial applications of critical importance as decision of document authenticity can also end disputes.

Deep learning, a sub field of Artificial Intelligence, is an effective solution for addressing the stated problem of forged signatures [9]. AI models can use neural networks to learn complex patterns in signature images and distinguish subtle differences between a genuine and forged sample with high accuracy. Convolutional Neural Networks (CNNs) are superior at extracting spatial features from images, whereas sophisticated methods such as attention mechanisms and Long Short-Term Memory (LSTM) units extend the model's capacity to pay attention to important areas and absorb sequential dependencies, respectively [10]. These capabilities position AI as the most suited for signature verification to produce trustworthy systems that can manage large datasets, generalize across diverse styles of signatures, and provide objective decisions in real time [11].

To solve the forged signature problem, we propose a novel deep learning model CALSigNet (CNN-Attention-LSTM Signature Network). CALSigNet is a combination of a CNN backbone with an attention mechanism and an LSTM unit that accurately distinguishes between forged and genuine signatures [12]. The research shows the model's superior performance with rigorous testing by kfold cross-validation, baseline CNN model comparison, and the model's real-world applicability in forensic science. This is achieved by leveraging the power of AI with the personalized approach of CALSigNet to enhance the ability to validate documents in legal and financial settings, thereby promoting the use of AI in forensic sciences [13].

II. RELATED WORKS

Artificial intelligence has emerged as a key factor of forensic sciences, helping to overcome the difficulties in

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processing intricate and massive data in criminal investigations. Studies identify AI potential in the progression of forensic methods in areas of blood pattern identification [14], crime scene reconstruction, digital forensics, and image analysis, while stressing the need for transparent and ethical AI mechanisms to minimize wrongful convictions caused by flawed scientific research or organizational negligence [15]. Also, research investigates AI applications in forensic image and audio examination, for example, the identification of crime tools and establishing causes of death, highlighting the need for high-quality information and model interpretability for consistent results [16]. In computer cybersecurity forensics, a Hybrid Intrusion Detection System (HIDS) integrates signature-based detection and AI-based anomaly detection utilizing Gradient Boosting and K-Nearest Neighbors (KNN) algorithms to reach 90.37% accuracy in detecting cyber threats while suppressing false positives [17].

Signature verification, an important forensic process, has relied on human methods in the past, that are subject to errors, initiating the use of AI for automation. One proposed banking application uses AI to scan signatures into images, draw out features, and utilize a neural network to identify them as genuine or counterfeit, enhancing dependability over human examination [18]. Another framework solves for misplaced signatures on student admission cards through AI-based identification of photographs and signatures in uploaded documents, proper placement, and avoidance of administrative mistakes [19]. Also, a signature recognition system utilizes Support Vector Machine (SVM) and Histogram of Oriented Gradients (HOG) using OpenCV and Qt for the GUI, where HOG extracts the feature vectors by reducing images, and SVM classifies them into different classes with user-specified precision [20].

Regardless of these developments, most current methods focus on static image analysis and ignore temporal dynamics that can be used to improve detection accuracy. Most methods do not capture spatial and sequential features of signatures thoroughly, restricting their use in detecting complex forgeries [21]. This gap prompts the current study, proposing CALSigNet, a new model that encompasses CNN, Attention, and LSTM to address such limitations by bringing together spatial feature extraction, region-based attention, and sequential pattern detection for better signature forgery identification [22].

Verification of signatures by AI systems in critical legal and financial settings creates ethical issues because incorrect classifications could cause severe problems including unfair court decisions and financial scams. The use of biased training data leads to misidentification of signatures particularly from specific demographic groups like elderly people with hand-writing trepidation thereby raising fairness problems that interfere with legitimate claims and business deals. Banking systems must exercise caution when storing signature data without consent because GDPR violations are likely to occur while processing sensitive personal client information. Legal proceedings and judicial acceptance become jeopardized by unexplained AI decision-making because transparency remains a barrier in this process. The deployment of these systems becomes more complex due to ethical problems that necessitate strong protective measures to create fair and secure and transparent verification processes.

III. METHODOLOGY

A. Dataset Collection and Preparation

We sourced our data from an open-source repository. The dataset contains 2152 signature image files that are divided between training and test sets [23]. We have 762 forged and 887 real signatures in the training set (a total of 1649 samples), and in the test set, we have 248 forged and 252 real signatures (a total of 500 samples) as depicted in Figure 1 and Figure 2. To train our models, we ran a preprocessing pipeline in Python that employed TensorFlow and Keras. Loaded images were transformed into grayscale, resized to the size of 128x128 pixels, and normalized to range [0, 1] by dividing pixel values by 255 [24], [25]. Preprocessing was carried out on files ending with .png, .jpg, or .jpeg in a case-insensitive fashion, handled errors, and generated training (X_train, y_train) and test (X_test, y_test) datasets of shapes (1649, 128, 128, 1) and (500, 128, 128, 1), respectively. For forged, labels were designated as 0 and genuine signatures as 1, with training data shuffled to avoid order bias [26].

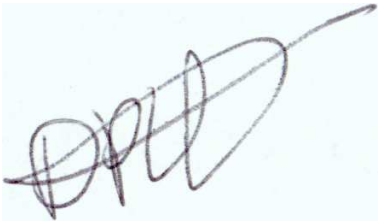


Fig. 1. Forged Signature image in the dataset



Fig. 2. Genuine Signature image in the dataset

B. CALSigNet Proposed Architecture

We developed a hybrid deep learning model, called CALSigNet to detect real and fake signatures, implemented via Keras in TensorFlow. The architecture starts with an input layer which can accept grayscale images of size (128, 128, 1), followed by a CNN backbone consisting of three convolution layers (32, 64, and 128 filters, 3x3 kernels, ReLU activation, same padding) and MaxPooling2D layers (2x2 pool size), downsampling the outputs to (16, 16, 128) [27]. Then, a MultiHeadAttention layer (4 heads, key dimension 128) concentrates on key signature attributes, for example, stroke patterns, whose output is flattened to a 128-size vector through GlobalAveragePooling2D [28].

The attention output is then reshaped to (1, 128) and input into an LSTM layer with 64 units, which strengthens feature aggregation with the ability to catch possible sequential

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dependencies despite the fixed image context [29]. The model finishes with a Dense output layer with sigmoid activation, producing a probability score (≥ 0.5 for real, < 0.5 for fake). Compiled with the Adam optimizer, binary cross-entropy loss, and accuracy as the metric, this architecture integrates CNN's spatial feature extraction, Attention's region-specific attention, and LSTM's sequence modelling to enhance forgery detection performance [30], [31].

C. Model Training and Validation

CALSigNet was trained on preprocessed training set (X_train, y_train) for 20 epochs with batch size 32 with validation on test set (X_test, y_test) to monitor generalization. Performance was measured using the evaluate method with test accuracy and loss, while Matplotlib plotted training and validation accuracy/loss curves to test convergence and overfitting [32]. To further analyse robustness, 5-fold cross-validation was applied to the combined dataset (X_full, y_full, 2149 samples) employing scikit-learn's KFold (n_splits=5, shuffle=True, random_state=42) [33]. For each fold, the split dataset contained roughly 1719 training and 430 validation samples; a new model instance that was trained for 10 epochs (batch size 32) on the training subset, and validation accuracy were monitored. Average accuracy and standard deviation across folds provided a reasonable performance estimate, demonstrating consistency under data variations [34].

D. Comparison with Baseline CNN Models and Signature Prediction

To measure the effectiveness of the new model, we benchmarked it against two baseline CNNs with the same training (X_train, y_train) and test (X_test, y_test) datasets [35]. The Basic CNN had two convolutional layers (32 and 64 filters, 3x3 kernels, ReLU) with 2x2 max-pooling, followed by a Flatten layer, a 128-unit Dense layer with ReLU, and a sigmoid output, representing a straightforward classification strategy. The ResNet-like CNN comprised a starting 32-filter convolutional layer, a residual block (two 32-filter layers and a skip connection) [36], max-pooling, a 64-unit Dense layer, and a sigmoid output, using residual learning for better gradient flow. Both were trained over 10 epochs (batch size 32), and their test accuracies were compared against the new model's, highlighting the superiority of its Attention and LSTM units in signature verification [37].

For actual application, we have built a function to preprocess a signature image (grayscale, 128x128, normalized) from a file path, resizes it to (1, 128, 128, 1), and passes it to the loaded CALSigNet model. It returns a probability score, labelling the signature as "Genuine" (≥ 0.5) or "Forged" (< 0.5) with confidence levels, plotted together with the image using Matplotlib.

IV. Results

A. Data Preparation and Model Overview

The dataset was preprocessed into train (1649 samples: 762 forged, 887 genuine) and test (500 samples: 248 fake, 252 real) sets, with dimensions of (1649, 128, 128, 1) and (500, 128, 128, 1) after grayscaling, resizing to 128x128, and normalization. These dimensions facilitate compatibility with deep learning models. CALSigNet model consisted of an input layer, three convolutional layers, a multi-head attention mechanism, an LSTM layer, and a dense output layer, with 405,953 trainable parameters [38]. This lean architecture, with most parameters in the convolutional and attention layers, is appropriate for signature verification, striking a balance between complexity and efficiency.

B. Model Training and Evaluation

The CALSigNet model was trained over 20 epochs (batch size of 32) using 1649 training samples validated on 500 test samples with clear improvements in performance. Accuracy to training improved from 53.45% (Epoch 1) to 95.82% (Epoch 20) with loss diminishing from 0.6911 to 0.1027, while increasing validation accuracy went from 50.40% to 99.60% by Epoch 19 (reaching loss at 0.0291) [39], slightly declining further to 0.0328 by Epoch 20. In Figure 3, accuracy was increased quickly, converging at Epoch 5 (training: 92.69%, validation: 97.80%) where curves plateau, with validation exceeding and flatlining above 98% post-Epoch 9; loss curves exhibited parallel training improvement but small validation fluctuation, with overfitting at Epoch 12 (validation accuracy falls to 94.00%, loss peaks to 0.2540) before reaching stability below 0.05. Test accuracy reached 99.60%, outperforming signature classification.[40].

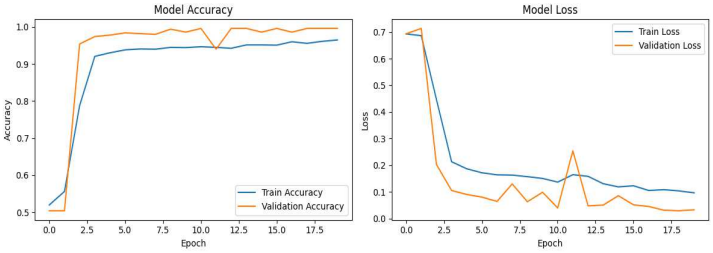


Fig. 3. Training and Validation Performance of CALSigNet Over 20 Epochs

Robustness was also confirmed using 5-fold cross-validation on the complete 2149-sample set (shape: 2149, 128, 128, 1). Output yielded a mean accuracy of 95.44% with standard deviation 0.0086 and fold accuracies of between 94.42% (Fold 2) and 96.74% (Fold 4), as summarized in Table 1. Such stability across splits is a guarantee of the stability and generalizability of the model to signature verification.

TABLE I. 5-FOLD CROSS-VALIDATION RESULTS FOR CALSIGNET MODEL.

Fold	Accuracy
1	0.9535
2	0.9442
3	0.9465
4	0.9674
5	0.9604

C. Data Preparation and Model Overview

The CALSigNet model was contrasted with two baseline CNNs—Basic CNN and ResNet-like CNN—on the test set, and results are presented in Table 2. The novel model performed a test accuracy of 99.60%, outperforming the Basic CNN (97.60%) and ResNet-like CNN (98.40%). This advantage comes from its attention mechanism, which focuses more intensely on important signature characteristics, and

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LSTM units, which learn sequential patterns, beating the less complex Basic CNN and the residual-improved ResNet-like CNN [41].

Model	Test Accuracy
Basic CNN	0.9760
ResNet-like CNN	0.9840
CALSigNet	0.9960

TABLE II. MODEL COMPARISON ON TEST SET.

In real-world testing, the trained model identified individual signatures to simulate forensic application. Signature 1 was labelled Forged with a confidence of 0.5398—the low score suggesting feature ambiguity—while Signature 2 was confidently labelled Genuine at 0.9962, demonstrating high reliability as shown in Figure 4. These results validate the model's forensic utility, performing well in straightforward cases and reasonably well in more challenging ones.

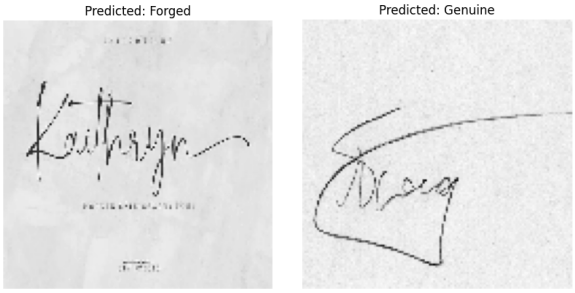


Fig. 4. Signature prediction results for sample images using CALSigNet

V. DISCUSSION

The high accuracy of CALSigNet (99.60%) implies its use as a forensic tool, but compatibility with current signature verification systems used in legal and financial institutions can increase its impact in real-world applications. An API integration of CALSigNet system into DocuSign platforms and financial institution workflows would enhance forgery detection alongside human review mechanisms. To fulfill industry requirements the system requires both compatibility tests along with confidence level refinement at 0.5398 levels or higher for increased market acceptance.

The feasibility of deploying CALSigNet must be evaluated through a complexity computational analysis. The 1.55 MB file size combined with training duration between 30 to 34 seconds per epoch shows efficient operation yet evaluation of standard hardware memory usage and inference time (between 0.431 to 0.991 seconds) will clarify system scalability. To make the Attention mechanism practical for forensic applications forensic operators must optimize its resource usage.

Web-based deployment of CALSigNet requires essential cybersecurity evaluations to prevent vulnerabilities which protect the system from threats. Security testing of data encryption together with access control systems and adversarial attack defenses would protect sensitive signature information by building trust and complying with forensic requirements in web-based applications.

VI. CONCLUSION

This study proposed CALSigNet, a novel deep learning network incorporating CNNs, an Attention unit, and an LSTM cell with an outstanding 99.60% test accuracy for detecting forged signatures, superior to a Basic CNN (97.60%) and a ResNet-inspired CNN (98.40%) and having a solid 95.44% (±0.0086) mean accuracy under five-fold cross-validation. Signature predictions emphasized its forensic potential with confidences up to 0.9962, but lesser ones such as 0.5398 in some scenarios show room for improvement. Leverage of CNNs for spatial information, Attention for region-based attention, and LSTM for sequential patterns, CALSigNet's lightweight 1.55 MB architecture offers a scalable, robust solution for authenticating legal and financial documents, anti-fraud, and trust building. Future improvements may include data augmentation, ensemble methods, bigger datasets, temporal stroke data, or real-time systems, in addition to testing against proficient, random, and simulated forgery methods, testing performance on larger datasets over various real-world use cases for scalability, and integrating interpretability methods such as SHAP and LIME to increase model transparency and user confidence, further solidifying it in AI-based forensic document authentication.

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